Database-backed visualization systems

CS524: Big Data Visualization & Analytics

Fabio Miranda

Database-backed visualization systems

- Modern interactive visualization stacks are *great at authoring* (declarative specs, linked views), but they often hit a wall when you try to handle:
  - Very large data.
  - Many coordinated components.
  - Fast interaction loops.
  - Interoperability across different view toolkits.

The Effects of Interactive Latency on Exploratory Visual Analysis

- Empirical, user-centered argument for why fast, scalable, linked interaction matter.
- Liu and Heer ran a controlled study adding **+500ms delay per interaction** into an otherwise interactive system, and measured both interaction logs and think-aloud reasoning.

Key finding: *+500ms caused measurable harm. Users interacted less, covered less of the dataset, and produced fewer observations / generalizations / hypotheses*

Liu and Heer, 2014

The Effects of Interactive Latency on Exploratory Visual Analysis

Zhicheng Liu and Jeffrey Heer

Abstract—To support effective exploration, it is often stated that interactive visualizations should provide rapid response times. However, the effects of interactive latency on the process and outcomes of exploratory visual analysis have not been systematically studied. We present an experiment measuring user behavior and knowledge discovery with interactive visualizations under varying latency conditions. We observe that an additional delay of 500ms incurs significant costs, decreasing user activity and data set coverage. Analyzing verbal data from think-aloud protocols, we find that increased latency reduces the rate at which users make observations, draw generalizations and generate hypotheses. Moreover, we note interaction effects in which initial exposure to higher latencies leads to subsequently reduced performance in a low-latency setting. Overall, increased latency causes users to shift exploration strategy, in turn affecting performance. We discuss how these results can inform the design of interactive analysis tools.

Index Terms—Interaction, latency, exploratory analysis, interactive visualization, scalability, user performance, verbal analysis

1 Introduction

One stated goal of interactive visualization is to enable data analysis at "rates resonant with the pace of human thought" [19, 20]. This goal entails two research directions: understanding the rate of cognitive activities in the context of visualization, and supporting these cognitive processes through appropriately designed and performant systems.

Latency is a central issue underlying these research problems. Due to the time required for query processing, data transfer, and rendering, data-intensive visualization systems incur delay. It is generally held that low latency leads to improved usability and better user experience. Unsurprisingly, multiple research efforts focus on reducing query and rendering latency for large datasets, which may include billions or more data points. Latencies in state-of-the-art systems can range from 20 milliseconds up to multiple seconds for a unit task [2, 28, 29].

Despite the shared goal of minimizing latency, the effects of interaction delays on user behavior and knowledge discovery with visualizations remain largely unevaluated. While previous research on the effects of interactive latency in puzzle solving [4, 17, 35, 36] and search [8] has shown that user behavior changes in response to millisecond-scale differences in latency, studies in other domains such as computer games report no significant effects [23, 39].

It is unclear to what degree these findings apply to exploratory visual analysis. Unlike problem-solving tasks or most computer games, exploratory visual analysis is open-ended and does not have a clear goal state. User interaction may be triggered by salient visual cues in the display, driven by *a priori* hypotheses, or carried out through exploratory browsing. The process is more spontaneous and is unconstrained by factors such as game rules.

How does latency affect user behavior and knowledge discovery in exploratory visual analysis? To answer this question, we conduct controlled experiments comparing two latency conditions, differing by 500ms per operation. We analyze data collected from both system logs and think-aloud protocols to test if (a) delay impacts interaction strategies and (b) lower latency leads to better analysis performance.

Our work makes the following contributions. First, we present the design and the results of a controlled study confirming that a 500ms difference can have significant impacts on visual analysis. Specifically, we find that (1) the additional delay results in reduced interaction and reduced dataset coverage during analysis; (2) the rate at which users make observations, draw generalizations and generate hypotheses (as determined using a think-aloud protocol) also declines due to the delay; and (3) initial exposure to delays can negatively impact overall performance even when the delay is removed in a later session. Second, we extend the insight-based evaluation methodology [37, 38] for comparative analysis of qualitative data regarding visualization use. We introduce a procedure for segmenting, coding and analyzing think-aloud protocols for visualization research. Our analysis contributes coding categories that are potentially applicable for future protocol analysis. Finally, our results show that the same delay has varying influences on different interactive operations. We discuss some implications of these findings for system design.

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2 Related Work

Our research draws on related work in scalable visualization systems, cognitive science and domain-specific investigations on the effects of interactive latency. We review relevant literature below.

2.1 Scalable Data Analysis Systems

Building low latency analysis systems has been a focus for many research projects and commercial systems, spanning both back-end and front-end engineering efforts. Spark [44, 45] supports fast in-memory cluster computing through read-only distributed datasets for machine learning tasks and interactive ad-hoc queries. Nanocubes [28] contribute a method to store and query multi-dimensional aggregated data at multiple levels of resolution in memory for visualization. Profiler [26] builds in-memory data cubes for query processing. Tableau's data engine [1] optimizes both in-memory stores and live connections to databases on disk. imMens [29] decomposes multi-dimensional data cubes into binned data tiles of reduced dimensionality and performs accelerated query processing and rendering on the GPU.

In cases where long-running queries are unavoidable, sampling and online aggregation [22] are often used to improve user experience. BlinkDB [2] builds multi-dimensional, multi-resolution samples and dynamically estimates a query's response time and error. With online aggregation [22], visualizations of estimated results are incrementally updated as a query progresses. Studies suggest that data analysts can interpret approximate results visualized as bar charts with error bars to make confident decisions [16].

2.2 Time Scales of Human Cognition

Decades of psychology research have produced evidence that different thought processes operate at varying speeds [25]. Newell [33] provides a framework outlining proposed time scales of human cognition. Relevant to studies of human-computer interaction are the cognitive (100 milliseconds to 10 seconds) and rational (minutes to hours) time bands. Within the cognitive band, Newell identifies three types of time constants: deliberate act, operation, and unit task. Card et al. [11] make similar distinctions using a different terminology. Table 1 summarizes these scales, exemplary actions, and the time ranges during which these actions occur.

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- Liu and Heer results imply that system performance is a first-class cognitive constraint. In other words: **if interactive updates drift into “half-a-second territory,” exploration quality drops.**
- Works in visualization are essentially a response to that:
  - Make interactive views so common interactions stay fast through optimization (e.g., caching, indexing).
  - immens
  - Nanocubes
  - TopKube
  - Time-Lattice
  - NeuralCubes
  - …

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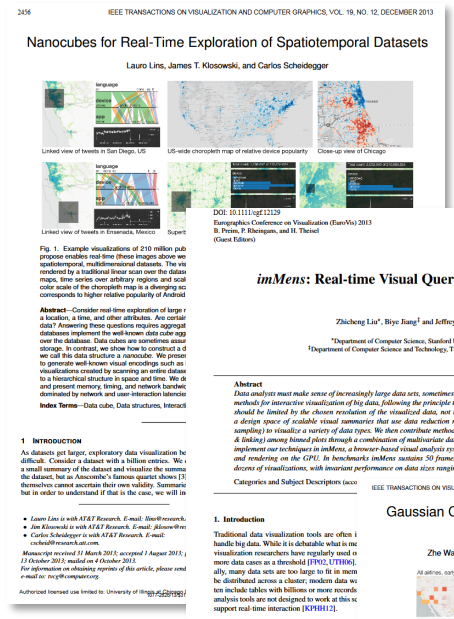
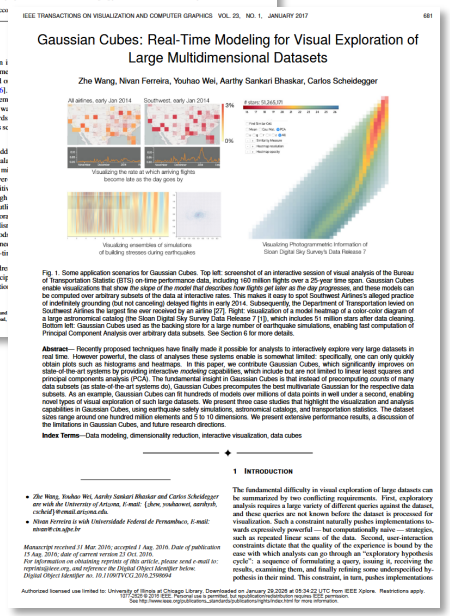
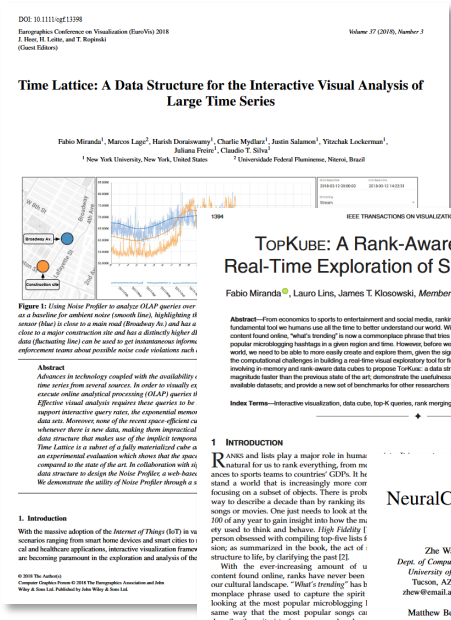


Lins et al., 2013

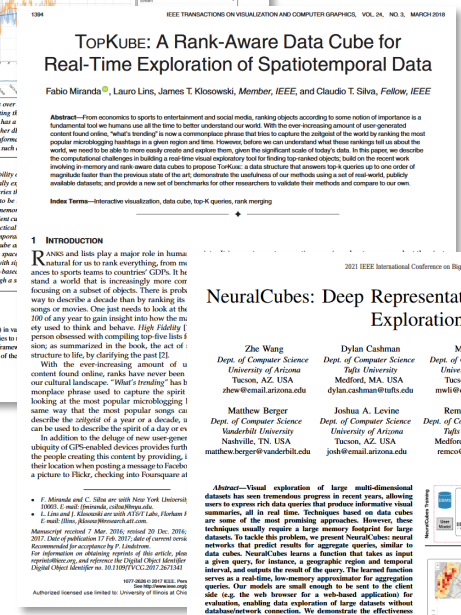
Liu et al., 2013



Wang et al., 2017



Miranda et al., 2018



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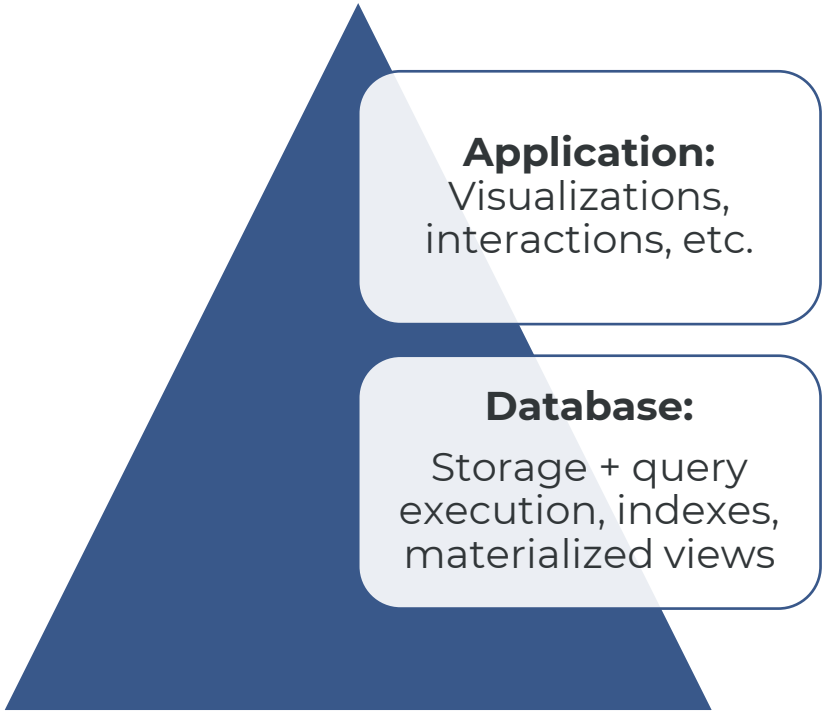
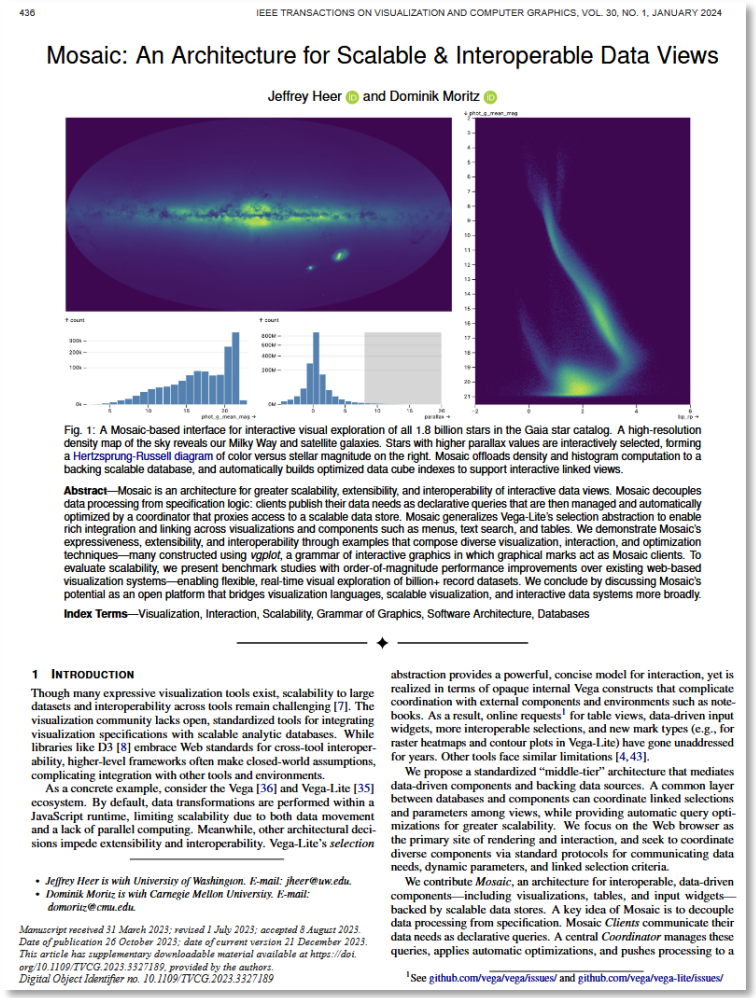


Heer and Moritz, 2024



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Jeffrey Heer and Dominik Moritz

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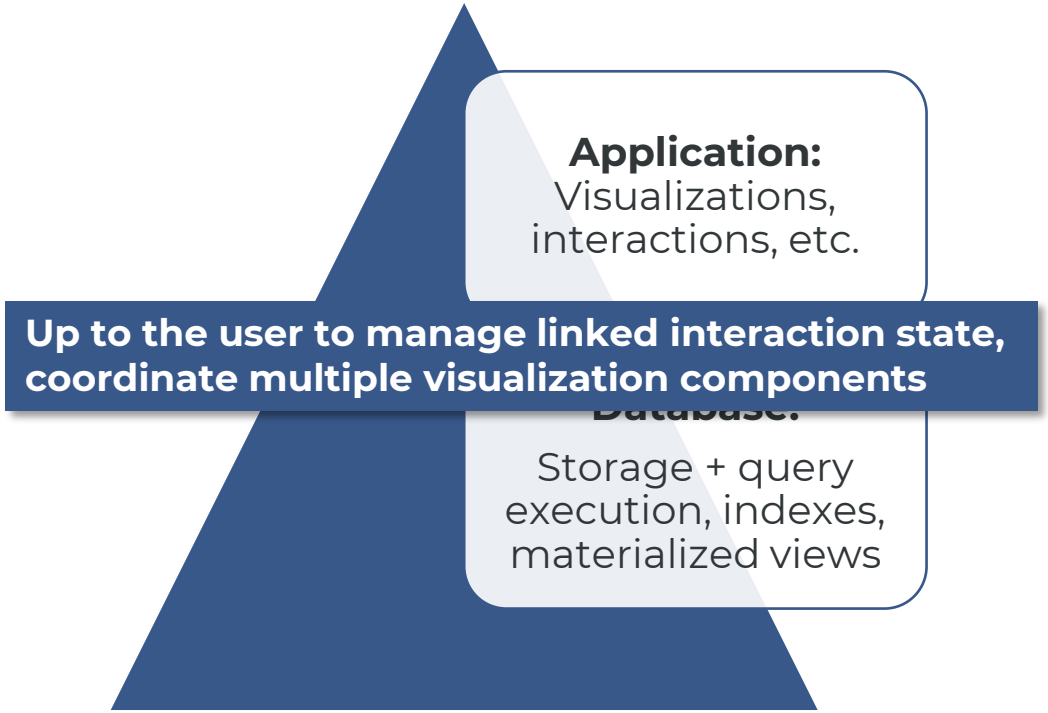

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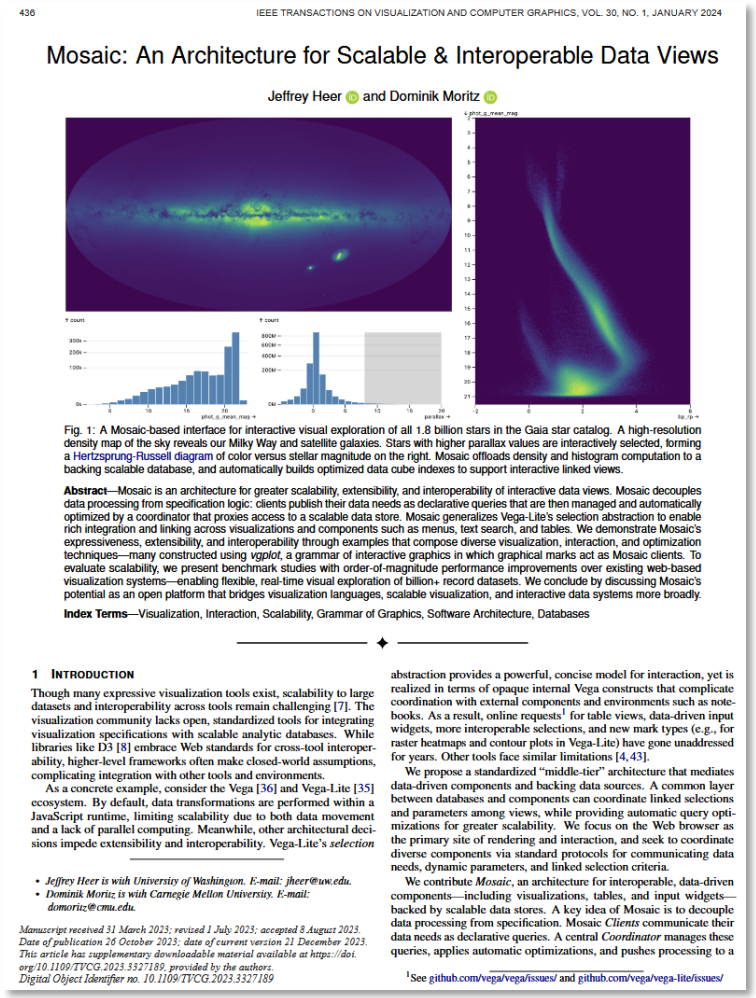

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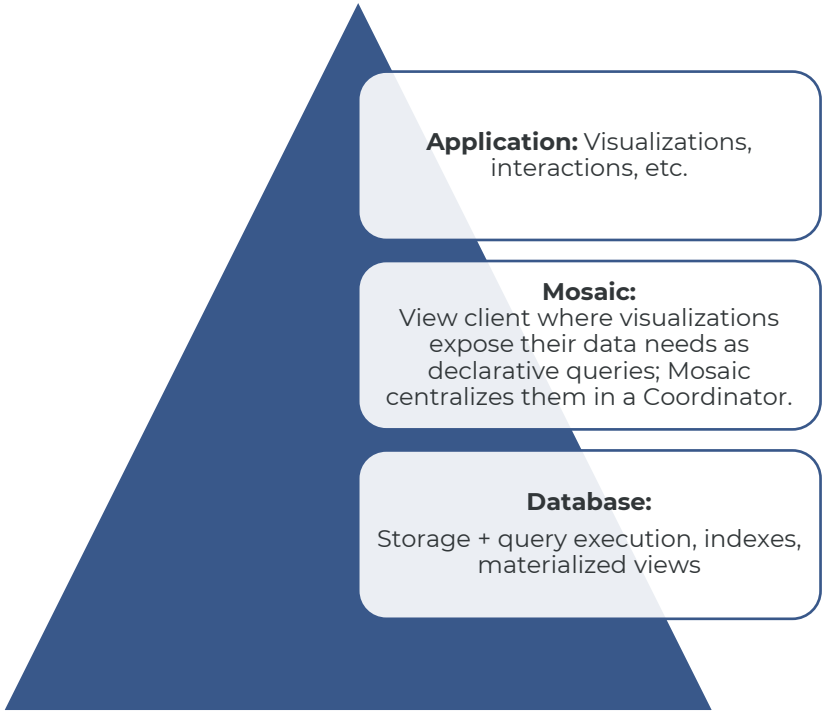

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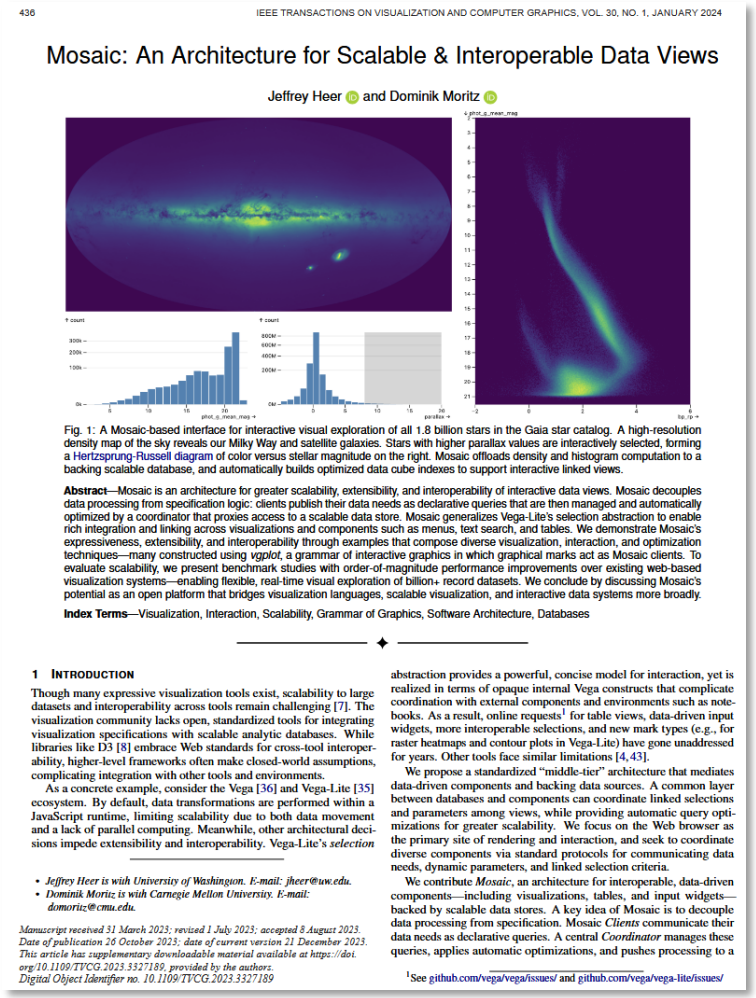

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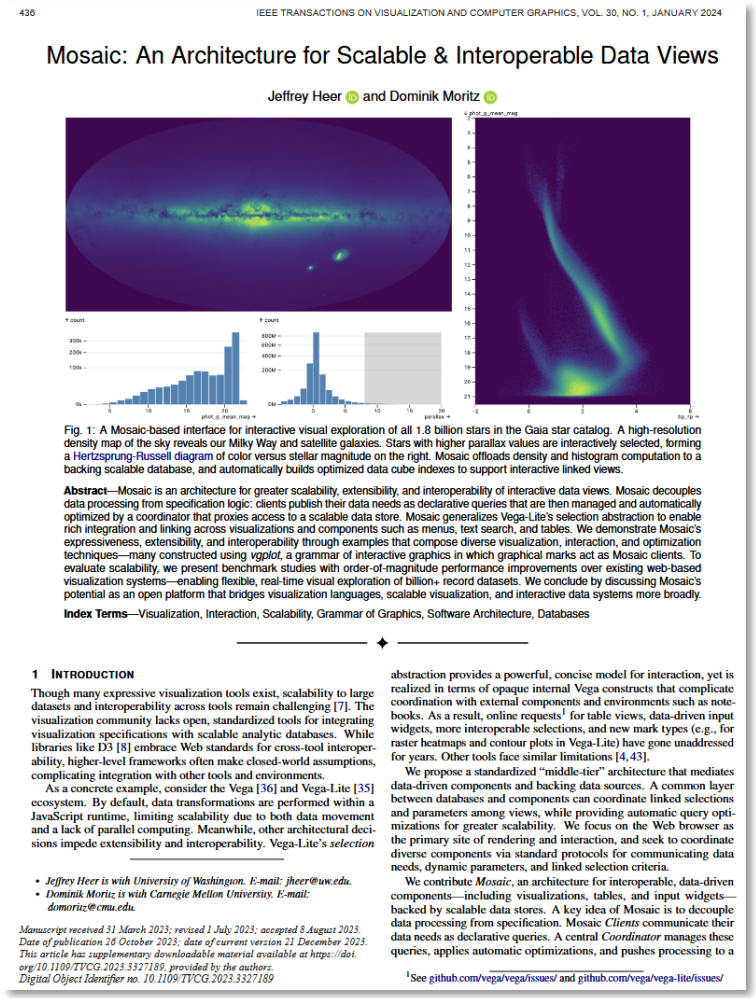
- DuckDB will answer queries.
- Mosaic will decide which queries exist, when they run, and how to optimize workloads globally.
  - Query caching
  - Query consolidation
  - Prefatching
  - Automatic data cube indexes

Heer and Moritz, 2024



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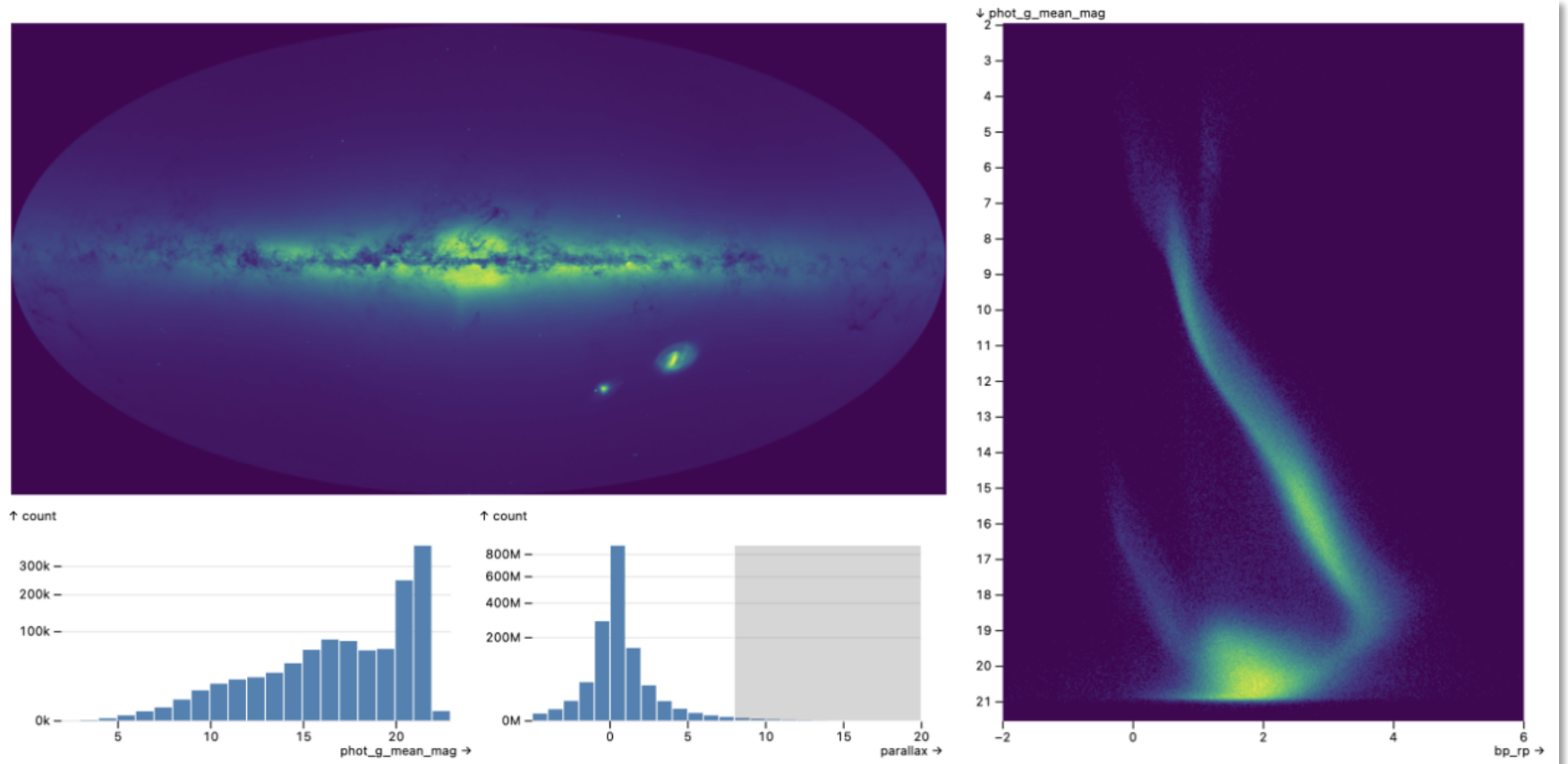

Fig. 1: A Mosaic-based interface for interactive visual exploration of all 1.8 billion stars in the Gaia star catalog. A high-resolution density map of the sky reveals our Milky Way and satellite galaxies. Stars with higher parallax values are interactively selected, forming a Hertzsprung-Russell diagram of color versus stellar magnitude on the right. Mosaic offloads density and histogram computation to a backing scalable database, and automatically builds optimized data cube indexes to support interactive linked views.

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- Query consolidation:
  - When several views request data around the same time, the Coordinator:
    1. Waits one animation frame;
    2. Collects incomings queries; and
    3. Merges those that hit the same table and the same GROUP BY dimensions into a single query.

View 1

```
SELECT month, AVG(price) AS avg_price
FROM sales
GROUP BY month;
```

View 2

```
SELECT month, SUM(price) AS total_sales
FROM sales
GROUP BY month;
```

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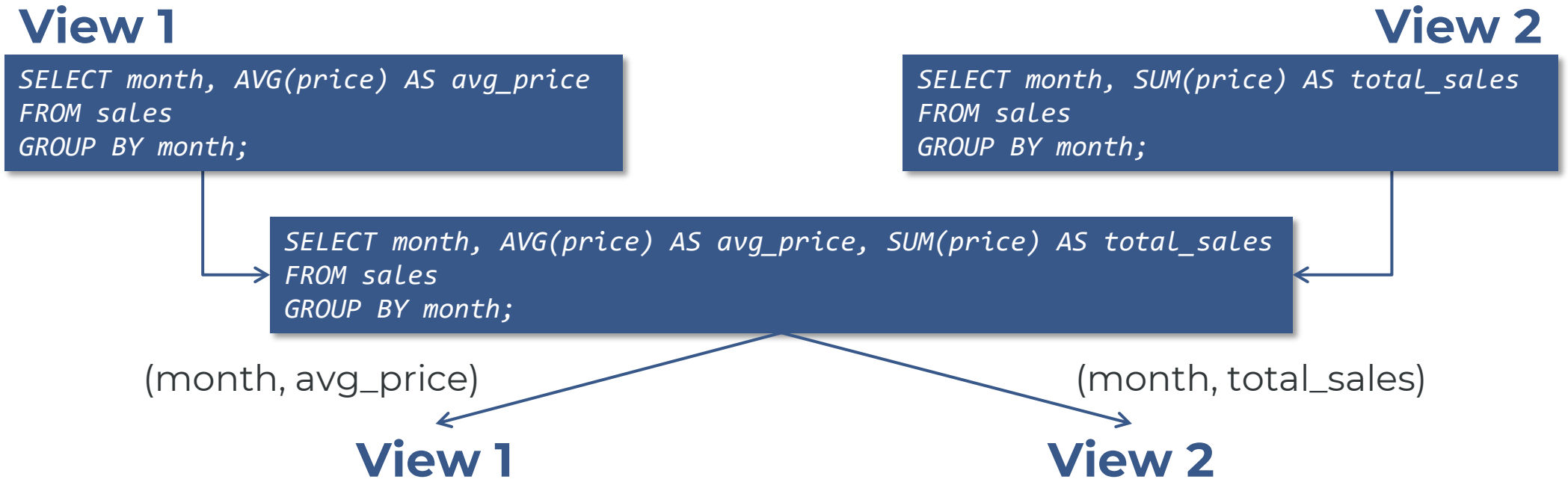

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- Data cube indexing:
  - Optimizes filtering of aggregated data, building indexes as small data cubes.
  - Without an index, every brush triggers: **filter raw rows → re-aggregate → redraw**
  - With a cube index, each brush triggers: **query a pre-aggregated table → sum a few bins → redraw**
  - Build once, reuse many times: data cubes are built based on actual workload.
  - Index size bounded by resolution, not by the number of data rows.
  - Key point: Mosaic doesn't compute the data cube of the entire dataset, but rather the smallest cube it needs for views at a bounded screen resolution.

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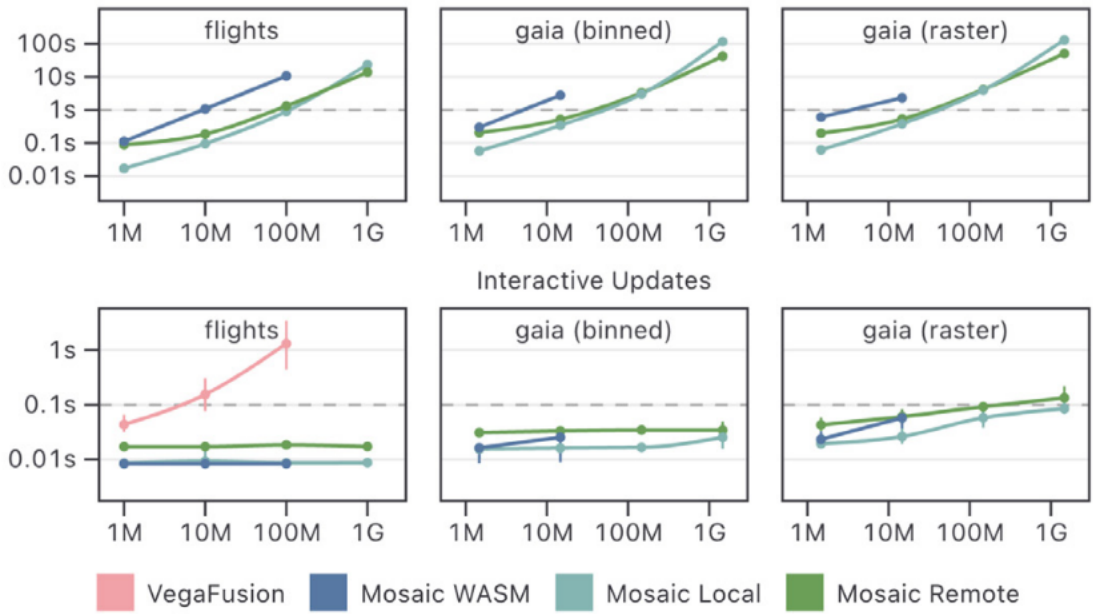

Fig. 17: Index construction (top) and interactive update performance (bottom). Median times and interquartile ranges are shown. Mosaic server configurations maintain interactive update rates at high data volumes, though start to degrade as indexes for raster data become denser.

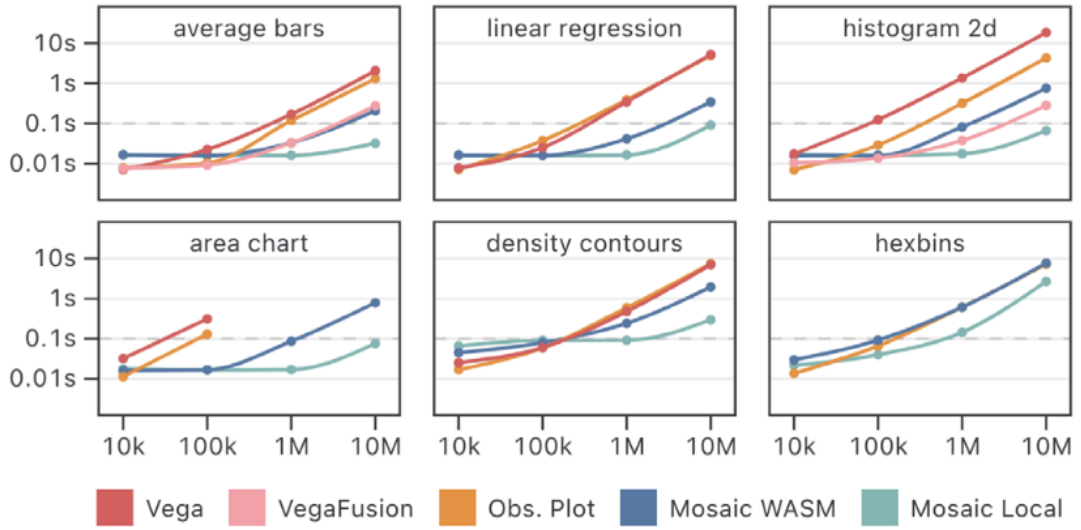

Fig. 16: Initial rendering performance. Median times are shown, interquartile ranges are smaller than the plotted dots. Mosaic provides order-of-magnitude performance improvements over Vega or Observable Plot for a range of visualizations. VegaFusion only optimizes the *average bars* and *2D histogram* conditions. Mosaic WASM does not scale as well as a local server, in part due to the lack of parallel processing.

Mosaic Selections: Managing and Optimizing User Selections for Scalable Data Visualization Systems

- Focuses on **linked interactions**.
- Interactive systems may have many *dependent* views to update.
- Mosaic Selections make selections explicit as structured objects:
  - Where did the selection come from?
  - Which views should it apply to?
- Formal structure allows for automatic optimizations.

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Mosaic Selections: Managing and Optimizing User Selections for Scalable Data Visualization Systems

Jeffrey Heer, Dominik Moritz, Ron Pechuk

Abstract—Though powerful tools for analysis and communication, interactive visualizations often fail to support real-time interaction with large datasets with millions or more records. To highlight and filter data, users indicate values or intervals of interest. Such *selections* may span multiple components, combine in complex ways, and require optimizations to ensure low-latency updates. We describe Mosaic Selections, a model for representing, managing, and optimizing user selections, in which one or more filter predicates are added to queries that request data for visualizations and input widgets. By analyzing both queries and selection predicates, Mosaic Selections enable *automatic optimizations*, including pre-aggregating data to rapidly compute selection updates. We contribute a formal description of our selection model and optimization methods, and their implementation in the open-source Mosaic architecture. Benchmark results demonstrate orders-of-magnitude latency improvements for selection-based optimizations over unoptimized queries and existing optimizers for the Vega language. The Mosaic Selection model provides infrastructure for flexible, interoperable filtering across multiple visualizations, alongside automatic optimizations to scale to millions and even billions of records.

Index Terms—scalable visualization, interactive selection, multiple coordinated views, brushing and linking, user interfaces

1 INTRODUCTION

To support effective analysis, interactive data systems should respond to user input at "rates resonant with the pace of human thought" [25, 34], as "milliseconds matter" [18] for user interface latency. To ensure perception of continuity and causality, foundational texts in Human-Computer Interaction [8] draw upon prior psychological studies to advocate interactive response rates under 100ms. Experiments [34, 59] find that adding 500ms of latency to interactive visualizations can alter analysts' strategies and degrade performance, leading to reduced exploration and hypothesis generation. However, not all interactive operations are equal, as the degree of latency sensitivity may vary [3,34]. User performance may be robust to a one second delay when creating a new chart, whereas the same latency for navigation operations such as panning and zooming can render an application unusable. In particular, updates for *linked selections*—also known as *brushing and linking* [5]—have been shown to be especially latency sensitive [34].

Though there are many expressive tools for visualization and data applications, most fail to scale to large datasets while preserving low-latency interactions [4, 58]. While researchers have developed methods for low-latency updates in response to dynamic user selections [2, 33, 35, 37, 41, 50, 51], most support only a limited subset of visualization types (such as scatter plots or time series) and interactions (such as panning and zooming). Moreover, many optimizations are not designed to work together in a single system, and require manual tuning or precomputation. Instead, we contend that interface designers and data analysts should be able to specify desired views and interactions at a high-level, and then benefit from automatic optimizations that, when possible, provide low-latency interaction ($\leq$ 100ms) over backing data volumes with millions or even billions of records.

We contribute a **formal model of user selections** in terms of *client views* that consume data, *interactors* that generate predicate clauses to select data subsets, and *selections* that manage clause sets to resolve view-specific filters. The selection model translates common operations performed in data analysis interfaces (using inputs such as menus, sliders, table views, and interactive visualizations) to filtering predicates applicable in database queries, and enables coordinated updates across multiple interface components. Unlike many prior formulations [10, 23, 46], our model flexibly supports *cross-filtering* configurations in which selections may apply to some interface views, but not others.

As datasets grow to millions or more rows, rendering all records becomes infeasible due to both perceptual and computational constraints [35]. Instead, it is common to bin, group, and aggregate data to achieve more scalable displays, ranging from basic histograms to high-resolution raster (pixel grid) views. Interactive updates must bin, filter and re-aggregate data, leading to prohibitively high latency if performed against a full dataset upon every update. To speed up these queries, **the selection model automatically optimizes updates using materialized views of pre-aggregated data.**[1] By analyzing client queries and a current active selection clause, the model determines appropriate dimensions and pre-aggregated measures to support efficient interactive updates. Querying the materialized views can make queries multiple orders of magnitude faster, enabling low-latency updates.

General multi-query optimization requires queries ahead of time and is NP-hard [29]. In contrast, queries for interactive visualization applications are predictable [35, 41]. The selection model leverages an application's structure to make multi-query optimization and answering queries using views [20] tractable. We detail a materialized view construction and query approach, which unlike prior methods [35, 41] supports aggregate operations beyond `COUNT` and `SUM`, and enables in-database computation and reuse across sessions and users.

We implemented the selection model and optimizations in Mosaic [24], an open-source architecture for linking databases and interactive views. By default Mosaic leverages DuckDB [44] for flexible deployment in web browsers, Jupyter notebooks, or using database servers. An earlier paper [24] presents the overall Mosaic architecture, describes included visualization and input widget components, and briefly introduces the selection model. We extend that work with a formal selection model, implementation details, and pre-aggregation optimizations with expanded aggregate function support. We also present performance benchmark results demonstrating significant improvements over naïve methods and existing systems, enabling a wider array of real-time interactions with large datasets.

In sum, this paper contributes (1) a formal model of user selections that bridges user interactions and backing databases; (2) optimizations that create and query pre-aggregated materialized views—applied *automatically* based on query and selection clause analysis—to accelerate aggregate queries for rapid interactive updates; and (3) performance benchmarks showing that Mosaic selections can reduce interactive latency by multiple orders of magnitude, outperforming existing systems.

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[1] A materialized view stores the result of a database query by name.

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Mosaic Selections: Managing and Optimizing User Selections for Scalable Data Visualization Systems

- Focuses on **linked interactions**.
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- Formal structure allows for automatic optimizations.

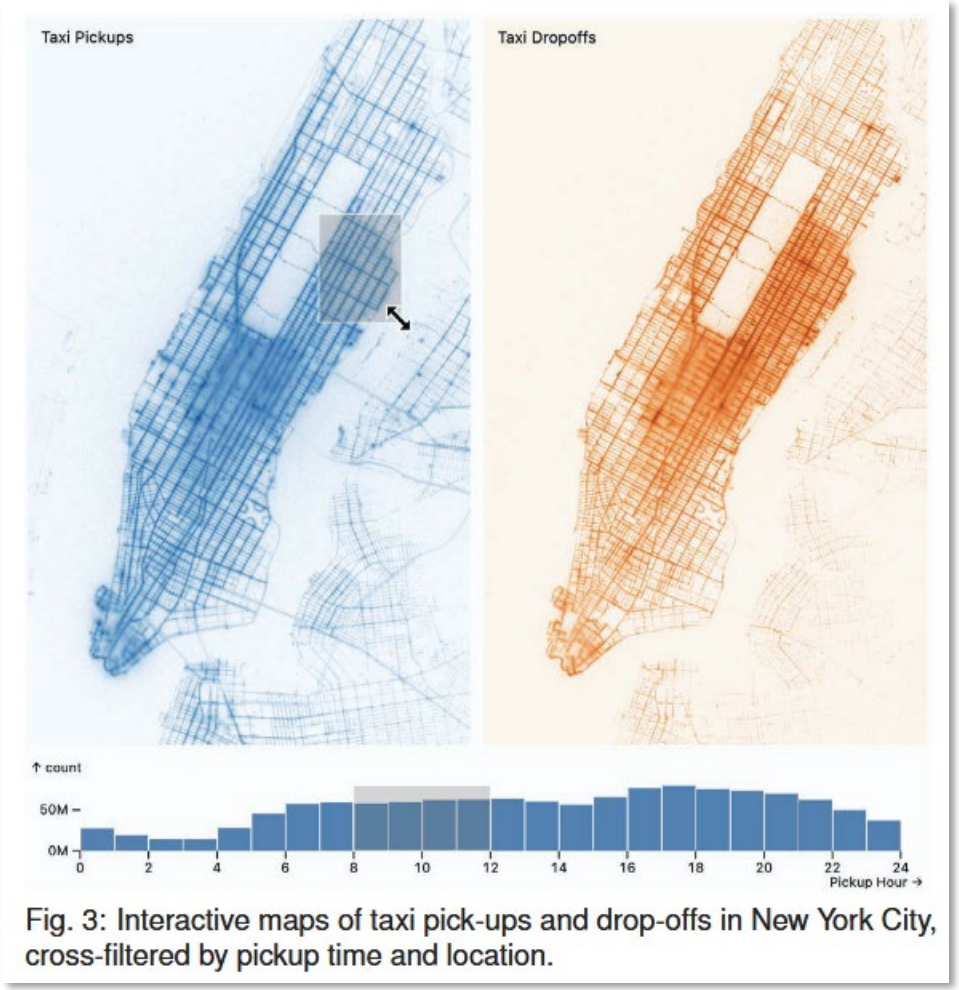

Fig. 3: Interactive maps of taxi pick-ups and drop-offs in New York City, cross-filtered by pickup time and location.

Mosaic Selections: Managing and Optimizing User Selections for Scalable Data Visualization Systems

- Expanded aggregate function support, with std. deviation, covariance, correlation.
- To automatically pre-aggregate data:

1. check if a query q_v filtered by active clause c_a performs aggregation amenable to optimization,
2. determine dimension columns by extracting any groupby values referenced in q_v, as well as dimensions for the possible values of the active clause c_a, and
3. create a materialized view with measure columns containing sufficient statistics for all aggregate query results.

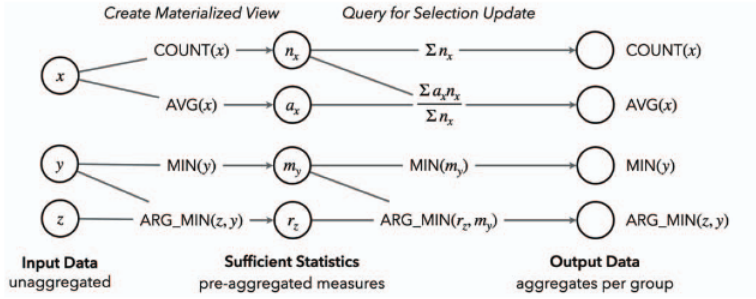

Fig. 8: Pre-aggregation and querying for univariate measures. Each sufficient statistic is included as a column in a materialized view, alongside grouping dimensions.

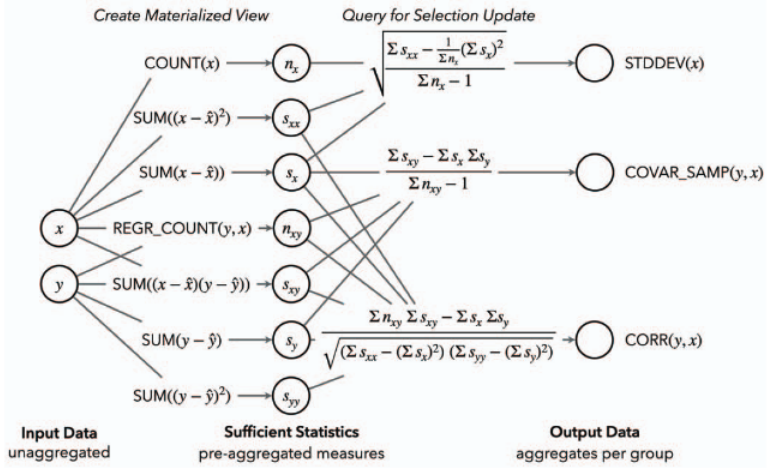

Fig. 9: Pre-aggregation and querying for standard deviation and bivariate measures. Each sufficient statistic is included as a column in a materialized view, alongside grouping dimensions. The symbol $\hat{x}$ indicates the average value of x across the full dataset; it is included to mean-center the data to prevent floating point error.

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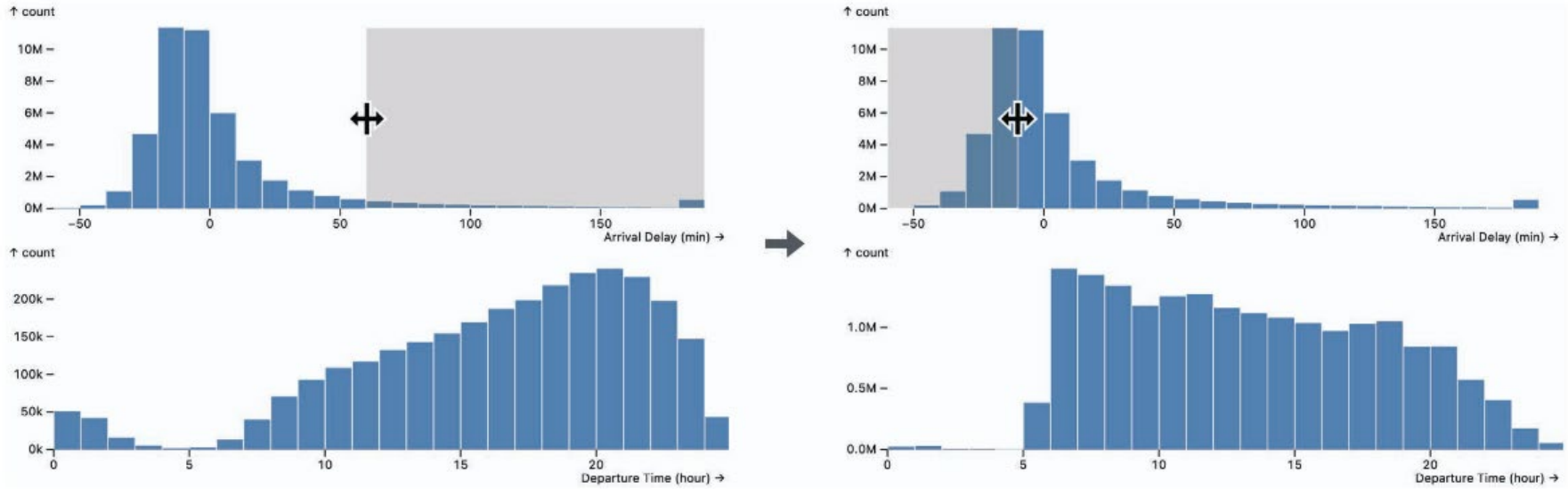

Fig. 1: Histograms of flight arrival data, interactively filtered by arrival delay intervals.

```
SELECT 24 * FLOOR(time / 24) AS x,
COUNT(*) AS y,
FLOOR($pixels * (delay - $min) / ($max - $min)) AS active
FROM flights
WHERE delay BETWEEN $delay_i AND $delay_j
GROUP BY x, active
```

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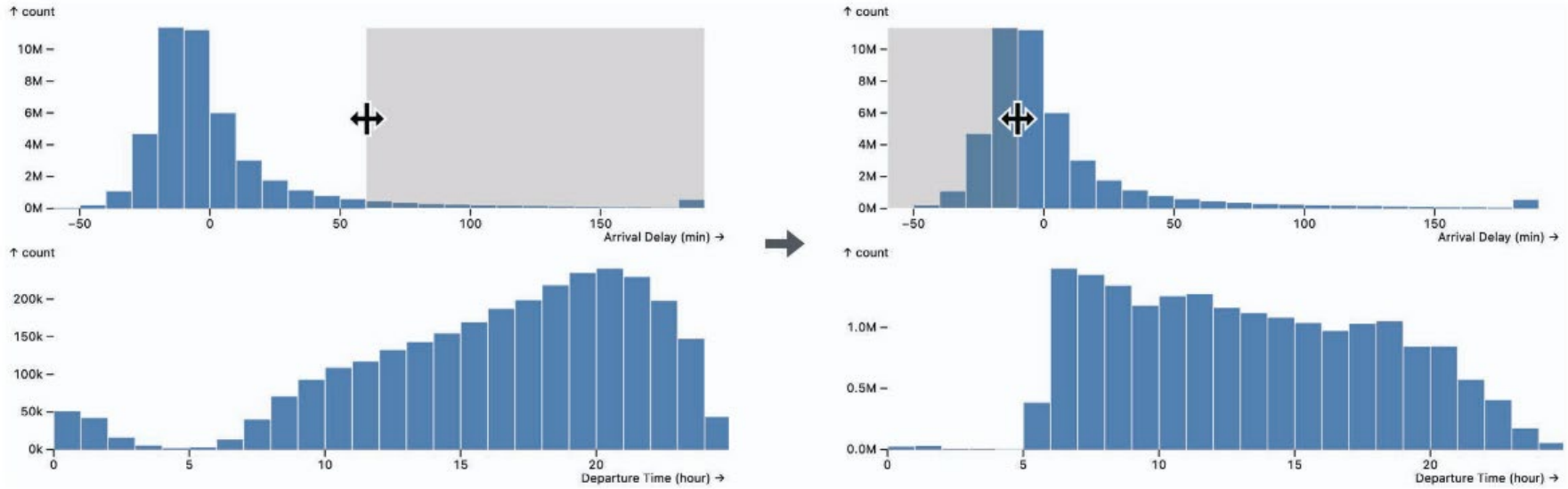

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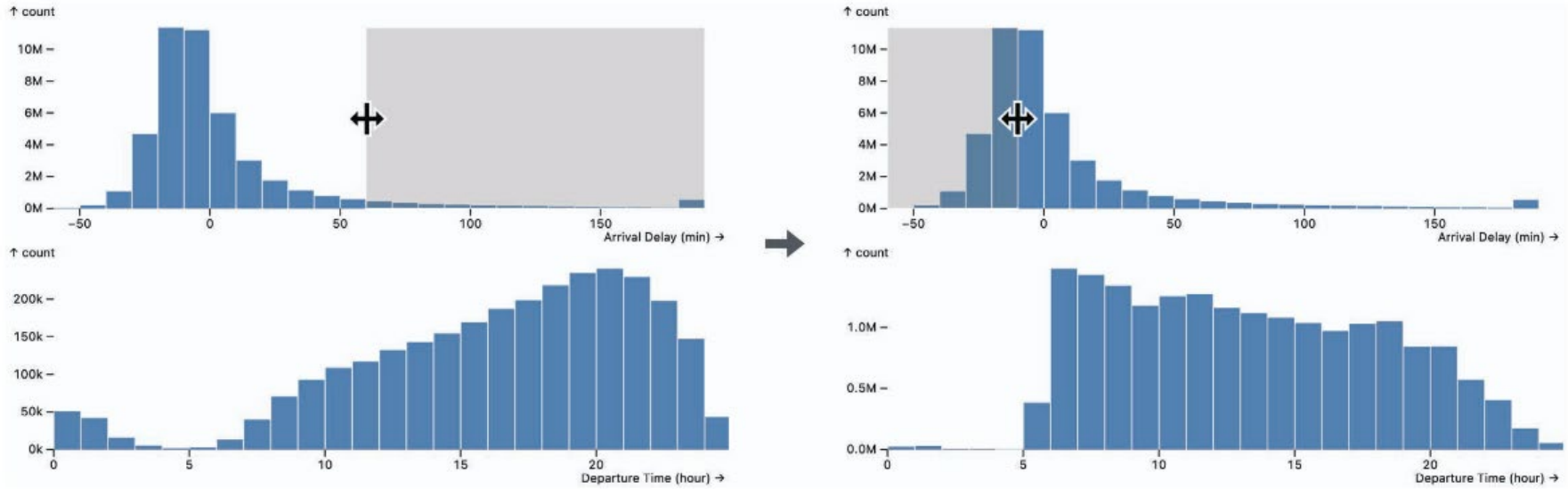

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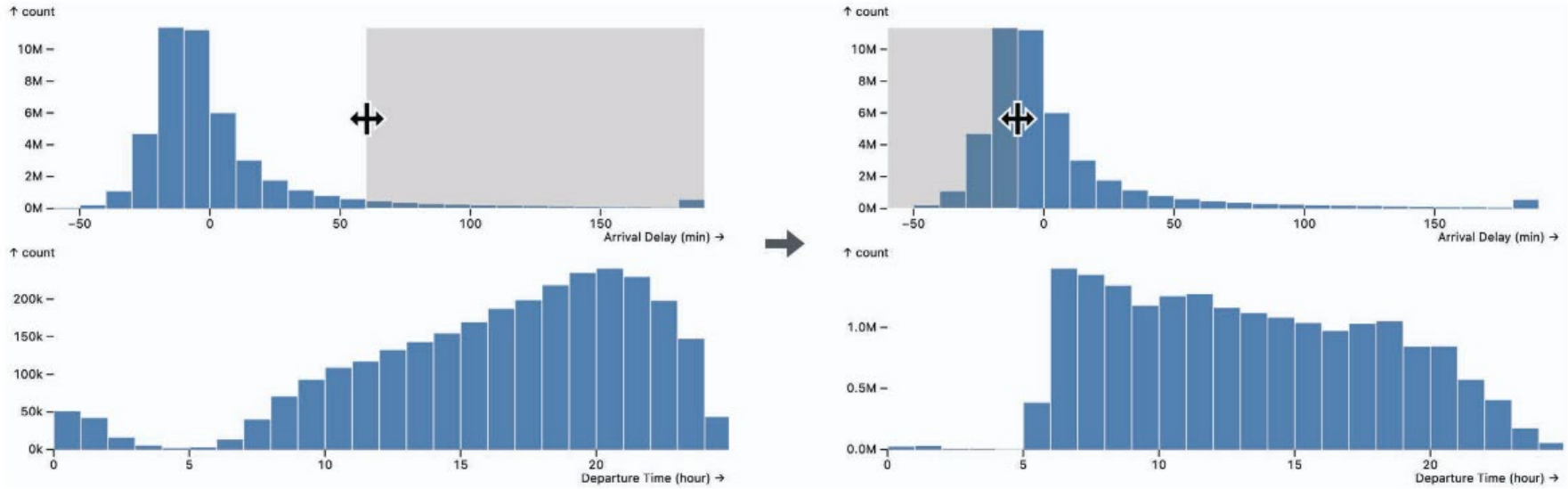

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